# Machine Learning

Benno Stein Theo Lettmann


© STEIN/LETTMANN 2005-2020

# Contents


I. Introduction

II. Machine Learning Basics

V. III. Linear Models

VI. IV. Neural Networks

VII. V. Support Vector Machines

III. VI. Decision Trees

IV. VII. Statistical Learning

VIII. Learning Theory

IX. Principles of Deep Learning

X. Ensemble Methods and Meta Learning

XI. Reinforcement Learning




©STEIN/LETTMANN 2005-2020

# Objectives

- understand and explain the basic concepts of machine learning
- understand formalized concepts and methods

- be able to implement concepts and methods in the form of algorithms
- be able to sensibly select, adapt, and apply relevant methods
- be able to educate oneself


© STEIN/LETTMANN 2005-2020

# Related Fields

1. Statistics [paradigms, models]
2. Mathematics

3. Information Retrieval [methods, algorithms]
4. Knowledge Processing
5. Heuristic Search

6. Decision Support Systems [applications]
7. Artificial Intelligence
8. Web Technology


© STEIN/LETTMANN 2005-2020

# Literature

Machine Learning:


- C.M. Bishop. [Interview (2018 MS Research)]
  *Pattern Recognition and Machine Learning*
  2nd edition, Springer 2006.

- L. Breiman, J.H. Friedman, R.A. Olshen, C.J. Stone.
  *Classification and Regression Trees*
  CRC Press reprint, 1998.

- N. Cristianini, J. Shawe-Taylor.
  *An Introduction to Support Vector Machines and Other Kernel-based Learning Methods*
  Cambridge University Press, 2000.

- T. Hastie, R. Tibshirani, J. Friedman.
  *The Elements of Statistical Learning*
  2nd edition, Springer, 2009. statweb.stanford.edu/∼hastie/ElemStatLearn/

- T. Mitchell.
  *Machine Learning*
  1st edition, McGraw-Hill, 1997. www.cs.cmu.edu/∼tom/mlbook.html

- V. Vapnik.
  *The Nature of Statistical Learning Theory*
  2nd edition, Springer 2000.



©STEIN/LETTMANN 2005-2020

# Software

Programming:

- Eclipse Foundation, Inc., Canada.
  *Eclipse SDK*
  Version 4.13. www.eclipse.org/downloads

Machine Learning:

- E. Frank, M. Hall, G. Holmes, M. Mayo, B. Pfahringer, T. Smith, I. Witten.
  *Weka Machine Learning Project*
  Version 3.8. www.cs.waikato.ac.nz/ml/weka

- *scikit-learn: Machine Learning in Python*
  Version 0.21 http://scikit-learn.org/stable/


© STEIN/LETTMANN 2005-2020

# Software

Statistics:

- R Development Core Team.
  *R*
  Version 3.6. www.r-project.org

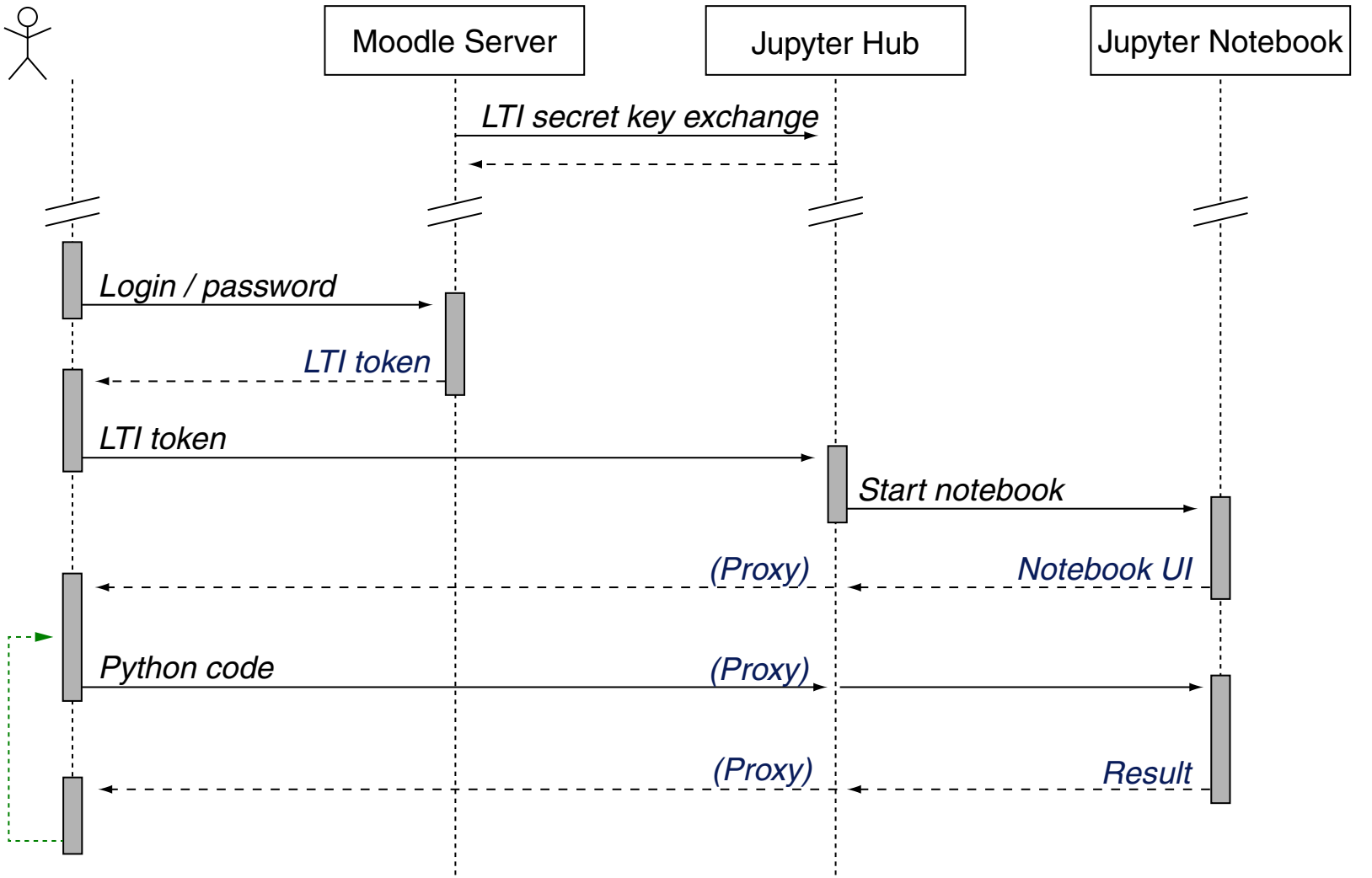
- E. Jones, T. Oliphant, P. Peterson and others.
  *SciPy*
  Version 1.17. www.scipy.org

- J.W. Eaton.
  *GNU Octave*
  Version 5.1. www.gnu.org/software/octave


© STEIN/LETTMANN 2005-2020

# Software

## Lab Class Setup

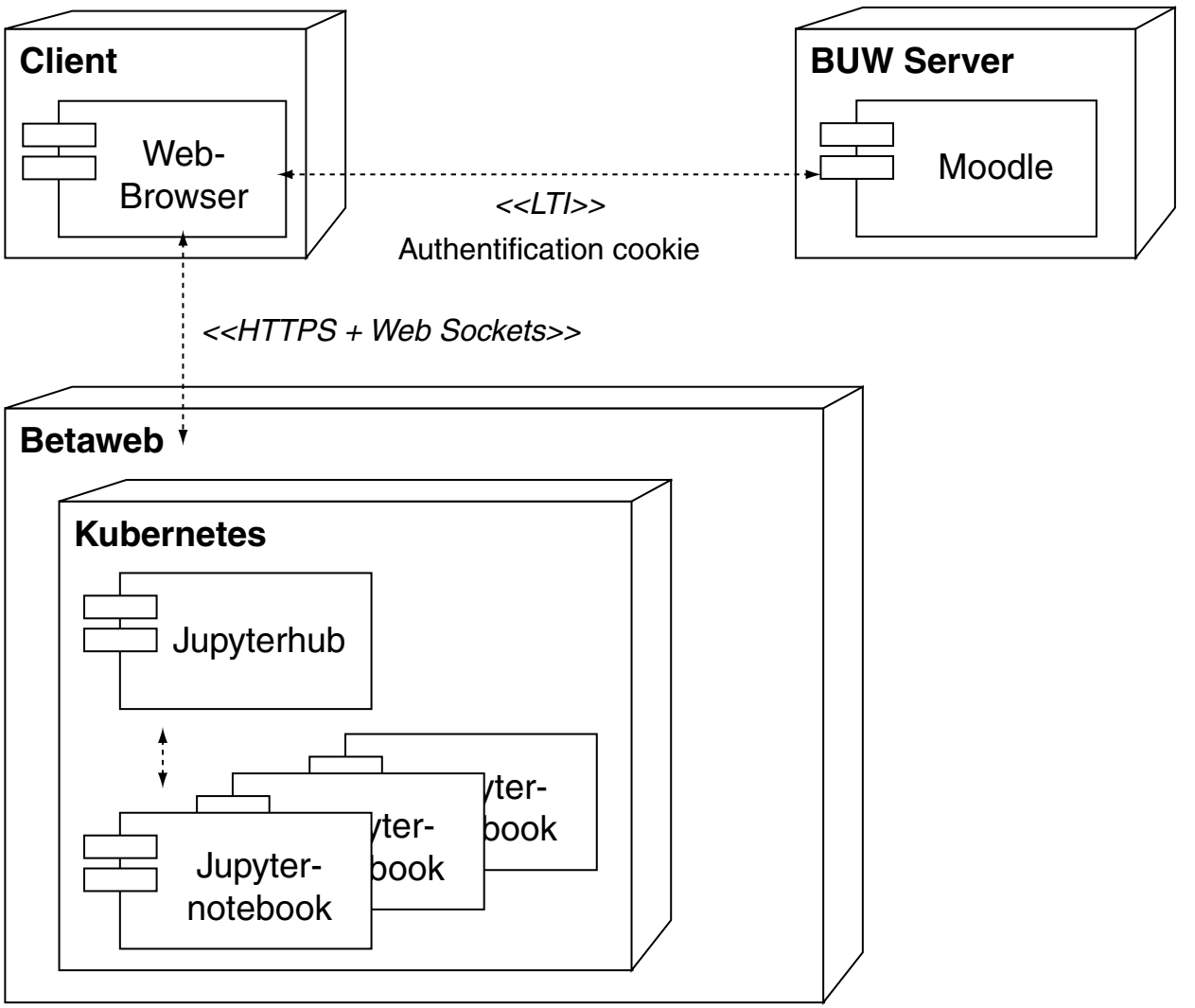




© STEIN/LETTMANN 2005-2020

# Software

## Lab Class Setup (continued)





© STEIN/LETTMANN 2005-2020